Supplementary Material

Dataset Description: China COVID-19 Surveillance Data

Language: Python
Data: China COVID-19 Surveillance Data
Date: March 2026

S1. Dataset Overview

The China dataset used in this study comprises daily recorded counts of three epidemiological indicators covering the COVID-19 pandemic in China from **1 April 2020 to 12 December 2022** (986 observations in total). The data were sourced from the *National Health Commission of China*. Given potential reporting delays and biases associated with asymptomatic case detection, the analysis focuses on confirmed symptomatic cases, which are considered to provide a more reliable and consistent basis for model fitting and prediction.

Property	Value
Data source	National Health Commission of China
Geographic scope	Mainland China (national aggregate)
Temporal resolution	Daily
Coverage period	1 April 2020 – 12 December 2022
Total observations (rows)	986
Number of features (columns)	3 (excluding date index)
Missing values	None (complete records across all features and dates)

S2. Feature Description

The dataset contains three epidemiological features. All counts are cumulative daily totals reported at the national level. The table below describes each feature, its unit of measurement, and its correspondence to the compartments of the TV-SIDARHE epidemic model.

Feature	Notation in Model	Unit	Description
Symptomatic Infections	R (Recognized)	Number of persons	Daily count of confirmed symptomatic COVID-19 cases with documented clinical presentation. Corresponds to the Recognized (R) compartment in the TV-SIDARHE model: detected symptomatic infected individuals. This variable is the primary target for model fitting and SPC monitoring.
Confirmed Cases	D (Diagnosed)	Number of persons	Daily count of all confirmed COVID-19 diagnoses, including asymptomatic detected individuals. Corresponds to the Diagnosed (D) compartment in the TV-SIDARHE model: detected asymptomatic infected individuals. Used jointly with Symptomatic Infections in Phase II SPC monitoring.
Deaths	E (Extinct)	Number of persons (cumulative)	Daily cumulative count of deaths attributed to COVID-19. Corresponds to the Extinct (E) compartment in the TV-SIDARHE model. This variable is used for initial condition specification and compartment conservation but is not a primary prediction or monitoring target.

S3. Descriptive Statistics

S3.1 Full Dataset

Descriptive statistics for the complete 986-day dataset are reported in Table S3. The wide range and high standard deviation of Symptomatic Infections and Confirmed Cases reflect the multi-wave epidemic trajectory across the observation period, which includes both the low-incidence plateau phase (April 2020 – early 2022) and the large-scale outbreak in late 2022.

Feature	Min	Max	Mean	Median	Std. Dev.
Symptomatic Infections (R)	97	396,094	19,734	494	62,471
Confirmed Cases (D)	55	43,266	3,583	861	7,581
Deaths (E, cumulative)	3,318	5,235	4,752	4,636	303

Note: The Deaths (E) column reports the cumulative national death toll; its relatively narrow range reflects the overall low case-fatality rate and stable reporting throughout the observation period. The large inter-quartile gap in Symptomatic Infections and Confirmed Cases is attributable to the sustained low-incidence period from April 2020 to early 2022 (median ≈ 494 and 861 respectively) followed by a rapid escalation in late 2022.

S3.2 First-Order Differenced Series

Because the raw Symptomatic Infections (R) and Confirmed Cases (D) series exhibit trend-stationarity during the Phase II monitoring period (as confirmed by ADF and KPSS tests reported in Table 5 of the main paper), first-order differencing is applied prior to SPC monitoring. Descriptive statistics for the differenced series over the full dataset are reported below for reference.

Feature	Min	Max	Mean	Std. Dev.
Δ Symptomatic Infections (ΔR)	−25,403	+23,948	233	4,072
Δ Confirmed Cases (ΔD)	−3,907	+3,453	35	452

S4. Data Partitioning and Usage

The dataset is partitioned into multiple non-overlapping or partially overlapping subsets according to the methodology described in the main paper. Three distinct analytical uses are identified: (i) TV-SIDARHE epidemic model fitting and prediction, (ii) SPC control chart Phase I baseline construction, and (iii) SPC control chart Phase II online monitoring and ensemble early-warning validation. Each partition is described in detail below.

S4.1 SPC Control Chart Partitions

S4.1.1 Phase I Training Set

The Phase I training set covers **1 April 2020 to 28 February 2022** (699 days) and is used to construct the in-control (IC) reference distribution for all control charts (K-SVDD, Time-Adaptive SVDD, D-SVM, and RTSVDD).

Attribute	Value
Period	1 April 2020 – 28 February 2022
Number of observations	699
Symptomatic Infections (R): range	97 – 1,104
Symptomatic Infections (R): mean ± std	456 ± 181
Confirmed Cases (D): range	55 – 3,530

Attribute	Value
Confirmed Cases (D): mean ± std	773 ± 570
Deaths (E): range	3,318 – 4,636
Role	Phase I in-control baseline for all four SPC control charts

S4.1.2 Reference Set Construction (S0)

From the 699-day Phase I training set, a representative reference set S_0 of **N_0 = 100 days** is constructed for SPC monitoring. Days are selected by identifying those whose Symptomatic Infections count falls within the 35th–50th percentile range of the Phase I distribution (a plateau-activity band of moderate epidemic intensity), and 100 days are then randomly sampled without replacement from this subset. A bootstrap procedure expands the Phase I sample to **N = 1,000** for offline control limit calibration.

Attribute	Value
Selection criterion	R within the 35th–50th percentile of Phase I R distribution
35th percentile of R (Phase I)	356 persons/day
50th percentile of R (Phase I)	391 persons/day
Candidate pool size (IQR band)	108 days
Reference set size (S_0)	100 days (random subsample without replacement)
Bootstrap expansion (offline training)	1,000 resampled observations

S4.1.3 Phase II Test / Monitoring Set

The Phase II monitoring set covers **1 March 2022 to 12 December 2022** (287 days). This period captures the major outbreak waves of 2022, including the large-scale resurgence in March–April, the moderate summer resurgence in August–September, and the post-zero-COVID transition outbreak in November–December.

Attribute	Value
Period	1 March 2022 – 12 December 2022
Number of observations	287
Symptomatic Infections (R): range	832 – 396,094
Symptomatic Infections (R): mean ± std	66,688 ± 101,254
Confirmed Cases (D): range	426 – 43,266
Confirmed Cases (D): mean ± std	10,426 ± 13,014

Attribute	Value
Deaths (E): range	4,636 – 5,235
Role	Phase II online monitoring for all four SPC control charts

S4.2 TV-SIDARHE Model Partitions

Two prediction scenarios are evaluated using the TV-SIDARHE model: a short-term and a long-term scenario. Both use sub-periods of the 2022 outbreak as training and test sets.

S4.2.1 Short-Term Prediction

Attribute	Training Set	Test Set
Period	25 February – 21 March 2022	22 March – 12 April 2022
Number of observations	25	22
R: range	880 – 20,155	22,000 – 258,374
R: mean ± std	6,203 ± 5,810	99,537 ± 72,098
D: range	2,431 – 23,138	21,826 – 29,306
D: mean ± std	8,434 ± 6,994	25,608 ± 2,453

S4.2.2 Long-Term Prediction

Attribute	Training Set	Test Set
Period	25 February – 18 May 2022	19 May – 20 June 2022
Number of observations	83	33
R: range	880 – 283,814	1,449 – 43,766
R: mean ± std	100,555 ± 82,891	12,125 ± 12,022
D: range	2,431 – 31,421	786 – 4,906
D: mean ± std	17,399 ± 9,832	2,056 ± 1,219

S4.3 Ensemble Framework Partitions (Early Warning Validation)

The ensemble early-warning experiment (Section 5.1.3 of the main paper) uses a 63-day Phase II monitoring window from **1 October to 2 December 2022**, partitioned into three consecutive stages:

Stage	Period	Days	Role	R: range	D: range
Stage 1	1 Oct – 31 Oct 2022	31	Simulation data; represents the post-summer stable period prior to the late-2022 outbreak. Not used for model training or prediction.	8,851 – 20,170	2,876 – 4,638
Stage 2	1 Nov – 21 Nov 2022	21	Training set for the TV-SIDARHE model in the ensemble experiment; covers the initial escalation phase of the late-2022 outbreak.	21,734 – 228,128	4,854 – 23,296
Stage 3	22 Nov – 2 Dec 2022	11	Predicted by the fitted TV-SIDARHE model; provides the forward trajectory of D and R for early-warning evaluation. Real observations for this period exist in the dataset and serve as ground truth.	247,193 – 396,094	24,630 – 39,358

Note: Stage 3 data are generated by the TV-SIDARHE model and are not drawn directly from the raw dataset during ensemble monitoring. The corresponding real observations in the dataset serve only as held-out ground truth for evaluating prediction accuracy (Table 6 of the main paper). The actual monitoring statistic for Stage 3 is computed from the model-predicted D and R trajectories.

S5. Preprocessing Steps

The following preprocessing steps were applied to the raw data before model fitting and control chart monitoring. All steps are described here to ensure transparency and reproducibility.

S5.1 Feature Scaling (Joint RobustScaler Normalisation)

All three features were normalised jointly using scikit-learn's `RobustScaler`, which centres each feature by its median and scales by its interquartile range (IQR). Fitting was performed on the combined array of training and test observations concatenated row-wise, ensuring that the same scaling parameters were applied consistently to both sets and preventing data leakage.

Step	Detail
Method	sklearn.preprocessing.RobustScaler
Centre	Per-feature median (robust to outliers)

Step	Detail
Scale	Per-feature interquartile range (Q75 − Q25)
Fit scope	Combined (training + test) to prevent leakage
Applied to	All three features prior to SVDD fitting and SPC monitoring

S5.2 First-Order Differencing for Stationarity

Prior to Phase II monitoring, the stationarity of the Symptomatic Infections (R) and Confirmed Cases (D) series was assessed using the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, both conducted at the 5% significance level with lag lengths selected via AIC (see Table 5 of the main paper). For the raw Phase II series (1 March – 12 December 2022), both ADF and KPSS null hypotheses were simultaneously rejected for D and R, indicating trend-stationarity. First-order differencing was applied to achieve strict stationarity, and both ΔD and ΔR passed both tests (ADF: $p < 0.001$; KPSS: $p = 0.100$). The differenced series were used as input to the control chart monitoring module.

Variable	ADF Statistic	p-value	KPSS Statistic	p-value	Stationarity
D (raw)	−4.761	< 0.001	0.157	0.041	Trend-stationary (rejected under both)
R (raw)	−4.171	0.005	0.161	0.038	Trend-stationary (rejected under both)
ΔD	−4.779	< 0.001	0.204	0.100	Strictly stationary
ΔR	−6.342	< 0.001	0.088	0.100	Strictly stationary

S5.3 Initial Condition Specification

For both TV-SIDARHE and SIDARTHE model fitting, the initial compartment sizes were set as follows, consistent with the observed epidemic state at the start of each training period. The total population was fixed at $\mathbf{N = 140{,}005 \times 10^4}$. The Susceptible (S) compartment was initialised as the residual population $\mathbf{S = N - I - D - A - R - H - E}$ (minus T for SIDARTHE) to satisfy the conservation identity at each starting point.

Compartment	Symbol	Value (short-term)	Value (long-term/ensemble)
Infected (undetected)	I	20,000	20,000
Diagnosed (detected asymptomatic)	D	880 (= R_0 from data)	21,734 (= R_0 from data)
Ailing (undetected symptomatic)	A	10,000	10,000

Compartment	Symbol	Value (short-term)	Value (long-term/ensemble)
Recognized (detected symptomatic)	R	2,431 (= R_0 from data)	4,854 (= R_0 from data)
Healed	H	101,786 (from data)	30,000
Extinct	E	4,636 (from data)	5,000
Threatened (SIDARTHE only)	T	100	100
Susceptible (derived)	S	N − all others	N − all others

S6. Consolidated Data Partition Summary

Table S6 consolidates all data partitions described in Sections S4 and S5, providing a single reference table for the complete experimental setup.

Partition	Period	n	Features Used	Purpose
Phase I (SPC)	1 Apr 2020 – 28 Feb 2022	699	ΔR, ΔD	IC baseline for all control charts; S_0 reference set construction
Reference set S_0	Sampled from Phase I (35th–50th pct of R)	100	ΔR, ΔD	SPC control limit calibration; Phase II adaptive reference
Bootstrap pool	Resampled from S_0	1,000	ΔR, ΔD	Offline expansion of Phase I sample for control limit estimation
Phase II (SPC monitoring)	1 Mar 2022 – 12 Dec 2022	287	ΔR, ΔD	Online monitoring for all four SPC control charts
TV-SIDARHE short-term train	25 Feb – 21 Mar 2022	25	R, D	TV-SIDARHE and SIDARTHE model fitting (short-term)
TV-SIDARHE short-term test	22 Mar – 12 Apr 2022	22	R, D	Short-term prediction evaluation (MAPE, R^2)
TV-SIDARHE long-term train	25 Feb – 18 May 2022	83	R, D	TV-SIDARHE and SIDARTHE model fitting (long-term)

Partition	Period	n	Features Used	Purpose
TV-SIDARHE long-term test	19 May – 20 Jun 2022	33	R, D	Long-term prediction evaluation (MAPE, R^2)
Ensemble Stage 1 (simulation)	1 Oct – 31 Oct 2022	31	R, D	Stable pre-outbreak simulation for ensemble framework context
Ensemble Stage 2 (model train)	1 Nov – 21 Nov 2022	21	R, D	TV-SIDARHE fitting for ensemble early-warning experiment
Ensemble Stage 3 (predicted)	22 Nov – 2 Dec 2022	11	R, D (model output)	TV-SIDARHE forecast serving as SPC input; real data used for ground-truth evaluation only